SQL PROJECT

ONLINE SHOPPING MANAGEMENT SYSTEM

Project Description

An **online shopping platform** where:

- Customers register
- Products are listed by category
- Customers place orders
- Orders generate invoices
- Payments are recorded

1️⃣ CUSTOMER Table

Purpose

Stores customer details.

Structure

Column	Data Type	Description
cust_id	VARCHAR(5)	Primary Key
f_name	VARCHAR(50)	First name
l_name	VARCHAR(50)	Last name
area	VARCHAR(10)	Area code
phone_no	VARCHAR(10)	Phone number

Query

CREATE TABLE customer (

```
    cust_id VARCHAR(5) PRIMARY KEY,
    f_name VARCHAR(50),
    l_name VARCHAR(50),
    area VARCHAR(10),
    phone_no VARCHAR(10)
);
```

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2 CATEGORY Table

Purpose

Stores product categories.

Structure

Column	Data Type	Description
category_id	VARCHAR(5)	Primary Key
category_name	VARCHAR(30)	Category name

Query

```
CREATE TABLE category (
    category_id VARCHAR(5) PRIMARY KEY,
    category_name VARCHAR(30)
);
```

---

3 PRODUCT Table

Purpose

Stores product details.

Structure

Column	Data Type	Description
product_id	VARCHAR(5)	Primary Key

product_name	VARCHAR(50)	Product name
category_id	VARCHAR(5)	Foreign Key
price	DECIMAL(8,2)	Product price
stock	INT	Available quantity

Query

```
CREATE TABLE product (
   product_id VARCHAR(5) PRIMARY KEY,
   product_name VARCHAR(50),
   category_id VARCHAR(5),
   price DECIMAL(8,2),
   stock INT,
   FOREIGN KEY (category_id) REFERENCES category(category_id)
);
```

---

4 ORDERS Table

Purpose

Stores order details placed by customers.

Structure

Column	Data Type	Description
order_id	VARCHAR(5)	Primary Key
cust_id	VARCHAR(5)	Foreign Key
order_date	DATE	Date of order
total_amount	DECIMAL(10,2)	Total order value

Query

```
CREATE TABLE orders (
   order_id VARCHAR(5) PRIMARY KEY,
   cust_id VARCHAR(5),
   order_date DATE,
```

```
    total_amount DECIMAL(10,2),
    FOREIGN KEY (cust_id) REFERENCES customer(cust_id)
);
```

5️⃣ ORDER_ITEMS Table

Purpose

Stores products included in each order.

Structure

Column	Data Type	Description
order_item_id	INT	Primary Key
order_id	VARCHAR(5)	Foreign Key
product_id	VARCHAR(5)	Foreign Key
quantity	INT	Quantity ordered
price	DECIMAL(8,2)	Product price

Query

```
CREATE TABLE order_items (
    order_item_id INT AUTO_INCREMENT PRIMARY KEY,
    order_id VARCHAR(5),
    product_id VARCHAR(5),
    quantity INT,
    price DECIMAL(8,2),
    FOREIGN KEY (order_id) REFERENCES orders(order_id),
    FOREIGN KEY (product_id) REFERENCES product(product_id)
);
```

6️⃣ INVOICE Table

Purpose

Stores invoice details for orders.

Structure

Column	Data Type	Description
inv_no	VARCHAR(5)	Primary Key
order_id	VARCHAR(5)	Foreign Key
inv_date	DATE	Invoice date
amount	DECIMAL(10,2)	Invoice amount

Query

```
CREATE TABLE invoice (
   inv_no VARCHAR(5) PRIMARY KEY,
   order_id VARCHAR(5),
   inv_date DATE,
   amount DECIMAL(10,2),
   FOREIGN KEY (order_id) REFERENCES orders(order_id)
);
```

---

7 PAYMENT Table

Purpose

Stores payment details.

Structure

Column	Data Type	Description
payment_id	VARCHAR(5)	Primary Key
inv_no	VARCHAR(5)	Foreign Key
payment_date	DATE	Payment date
payment_mode	VARCHAR(20)	Mode (UPI/Card)
payment_status	VARCHAR(15)	Success / Failed

Query

```
CREATE TABLE payment (
   payment_id VARCHAR(5) PRIMARY KEY,
   inv_no VARCHAR(5),
   payment_date DATE,
   payment_mode VARCHAR(20),
   payment_status VARCHAR(15),
   FOREIGN KEY (inv_no) REFERENCES invoice(inv_no)
);
```

SQL Query Questions

Online Shopping Management System

CUSTOMER TABLE–Based

1. Display all records from the CUSTOMER table.
2. Find the first name and area of customer with cust_id = 'C03'.
3. List names and phone numbers of all customers.
4. Count the total number of customers.
5. Find customers staying in area 'DA', 'MU', or 'GH'.
6. Change the area of customer 'C01' to 'JK'.
7. Find customers whose first name starts with the letter 'P'.
8. Find customers whose area has 'A' as the second character.
9. Update the phone number of customer 'Rajesh' to 567889.
10. Delete customer with cust_id = 'C09'.

CATEGORY & PRODUCT TABLE–Based

11. Display all product categories.
12. List all products with price greater than 150.
13. Find products priced between 100 and 180.
14. Display product name and price of all products.
15. Count the number of products in each category.
16. Find the maximum and minimum product price.

17. Calculate the average price of all products.
18. Find products belonging to 'Electronics' or 'Clothing' category.
19. Increase the price of product 'Laptop' to 55000.
20. List products sorted by product name in ascending order.

ORDERS TABLE–Based

21. Display all orders placed by customer 'C02'.
22. Count total number of orders.
23. Find orders placed before August.
24. Display order_id and order_date of all orders.
25. Change the total amount of order 'O05' to 12000.

ORDER_ITEMS TABLE–Based

26. Display all order items.
27. Find total quantity ordered for each product.
28. Find products ordered more than 3 times.
29. Calculate total price per order using order_items table.
30. Find products that were never ordered.

INVOICE TABLE–Based

31. Display all invoices.
32. Find invoice details for order_id 'O03'.
33. Delete invoice with inv_no = 'I08'.
34. Change invoice date of inv_no 'I07' to 16-08-2023.
35. Find invoices generated between two given dates.

PAYMENT TABLE–Based

36. Display all payment records.
37. Find payments with payment status 'FAILED'.
38. Count number of successful payments.
39. Find payments made using 'UPI'.
40. Delete payment record with payment_id 'P09'.

JOIN & AGGREGATION–Based

41. Display customer names along with their order IDs.
42. List product names and quantities ordered by each customer.
43. Find total amount spent by each customer.
44. Display invoice number and customer name.
45. Find customers who have placed at least one order.
46. Find customers who have not placed any orders.
47. Display product name and category name for all products.
48. Find total sales amount for each category.
49. Find the customer who has spent the maximum amount.
50. Display product names ordered by customers from area 'DA'.